CSE 315

Microprocessors, Microcontrollers, and Embedded Systems

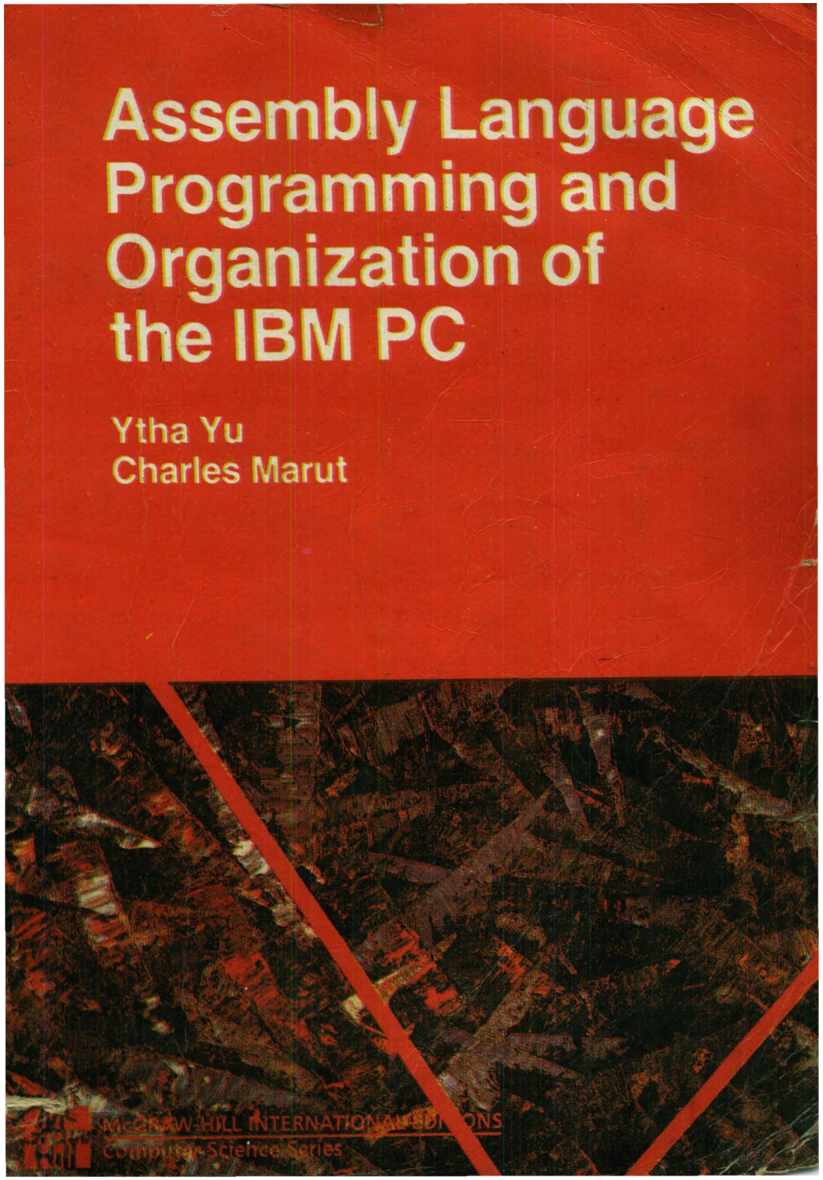
Information

- Course teachers
  - A. K. M. Mehedi Hasan Elin – MHE
  - Md. Emamul Haque Pranta – EHP
- Classes
  - 3 lectures per week
  - First 2 weeks (MHE): Assembly language
  - Middle 7 weeks (EHP): Microcontroller
  - Last 5 weeks (MHE): Microprocessor

CSE 316 Outline

- Microprocessors, Microcontrollers, and Embedded Systems Sessional
  - Assignments on assembly language
  - Assignments on microcontrollers (ATmega32)
  - Project on embedded systems

Textbook for Assembly Language



Textbook for Microprocessor

محجوزة طلبة الدراسة
(عند موظف المكتبة)

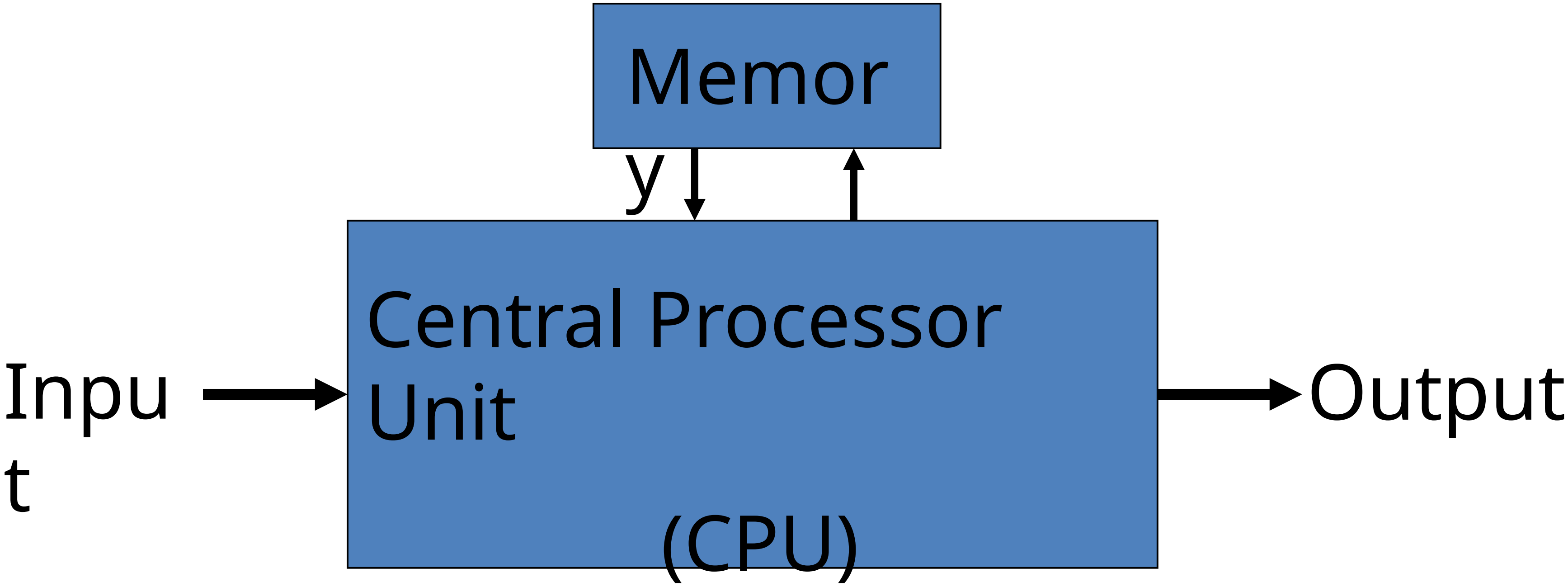
MICROPROCESSORS
AND
MICROCOMPUTER-BASED
SYSTEM DESIGN

1990

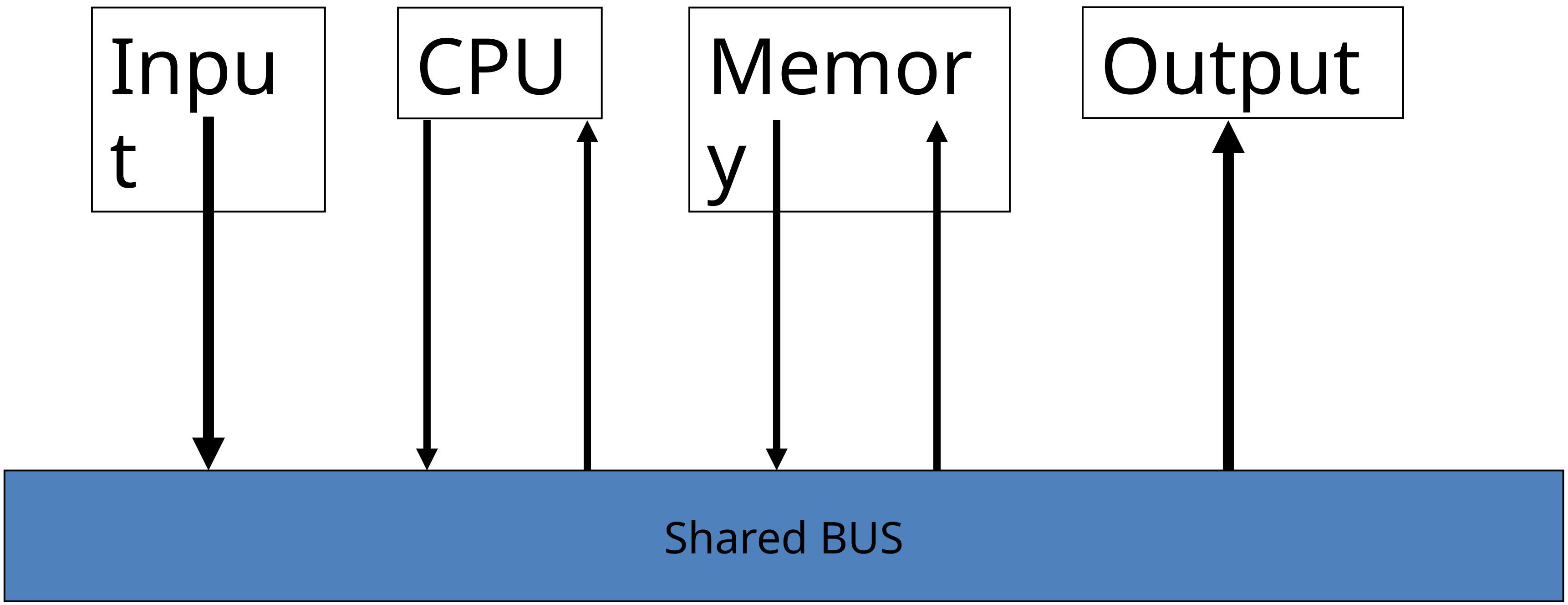
Mohamed Rafiquzzaman, Ph.D.
Professor
California State Polytechnic University
Pomona, California
and
Adjunct Professor
University of Southern California
Los Angeles, California

UNIVERSAL BOOK STALL
NEW DELHI

A Typical Computer System
Early days



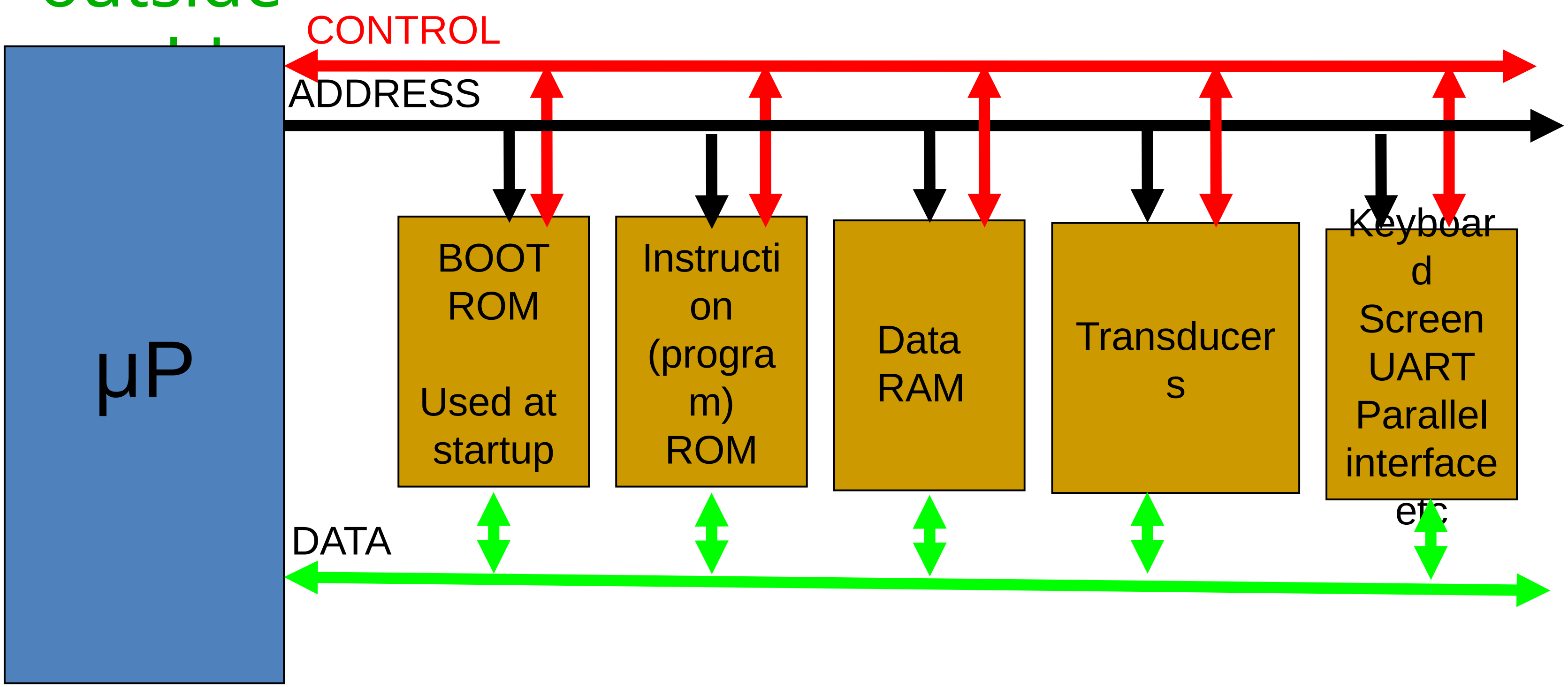
A Typical Computer System Nowadays



Microprocessor

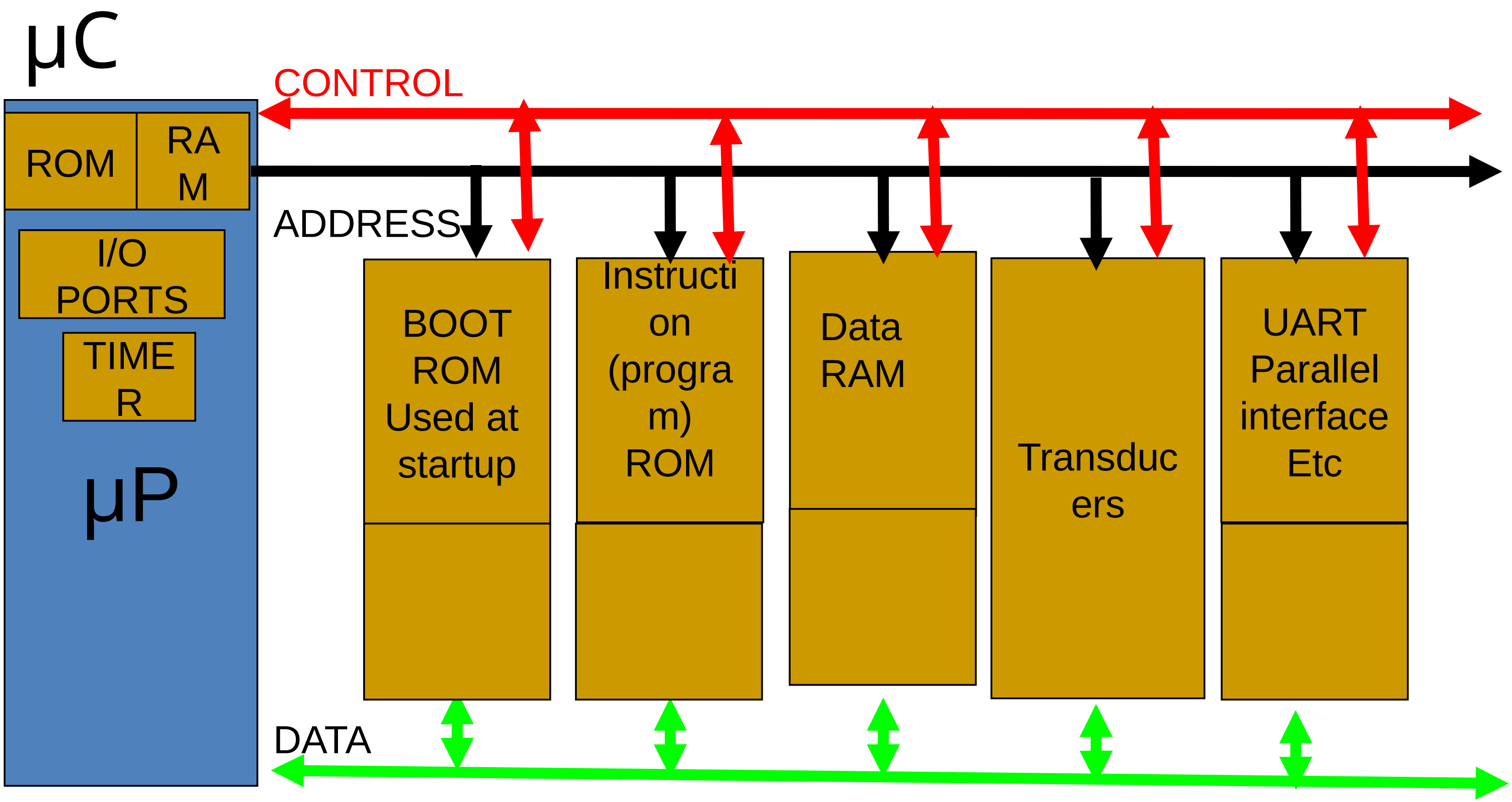
Microprocessor, by-itself, is completely useless

must have external peripherals to interact with outside



Many chips on mother board

MicroCONTROLLER



Microcontroller – put a limited amount of most commonly used resources "inside" the chip – a "limited" amount is often "enough" for many applications

Uses of µC

µP vs µC

	µP	µC
Structure	CPU	CPU, memory, I/O
Uses	Computer system	Embedded system
Purpose	General	Specific
Cost	High	Low
No. instructions supported	Large	Fewer
Speed	Fast	Slower
Power consumption	High	Low
Architecture	von Neumann	Harvard

Embedded systems

- An embedded system is a computer system that has a dedicated function within a larger mechanical or electrical system
- Usually based on microcontrollers
  - Complex ones can have microprocessors too
- Examples
  - Automobiles and airplanes
  - Medical devices
  - Home appliances

Embedded systems